

IDENTIFYING THE MOLECULAR BASIS OF THE AUTOINHIBITION RELIEF MECHANISM OF THE DISEASE-RELATED LIPID FLIPPASE ATP8B1

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ABSTRACT

P4-ATPases, also known as flippases, are eukaryotic membrane proteins capable of transporting phospholipids from the exoplasmic to the cytosolic leaflet of membranes using energy from ATP hydrolysis. This lipid transfer is crucial to create and maintain membrane lipid asymmetry, which is essential for cellular processes such as membrane trafficking, signal transduction, and resistance to mechanical and chemical stress. The human genome encodes 14 P4-ATPases, which form heterodimers with a CDC50 chaperone protein. Their physiological significance is underscored by the fact that approximately half of them have been linked to critical diseases. In particular, inactivating mutations in the flippases ATP8B1 are associated with progressive familial intrahepatic cholestasis [1]. Despite their importance, the precise molecular mechanisms behind flippase activation and their roles in disease remain poorly understood.

Here, we report progress made toward elucidating the post-translational modifications and protein partners, regulating ATP8B1. Previous work has revealed key insights into ATP8B1 regulation, including a strict autoinhibition mechanism mediated by its N- and C-terminal tails, which lock the flippase into an inactive conformation and prevent lipid translocation [1]. To investigate how this autoinhibition mechanism is relieved in vivo, the N- and C-terminal tails were expressed as eGFP-fusions peptides in *E. coli* and immobilized on anti-eGFP nanobody resin for pull-down experiments, in order to identify binding partners and/or phosphorylation by mass spectrometry. Peptides were incubated with liver-derived HepG2 cells lysate to capture native phosphorylation events and interacting proteins. In parallel, full-length ATP8B1-eGFP expression was achieved in HepG2 cells following a lentiviral transduction, providing a more native cellular context.

REFERENCES

- 1 Dieudonné et al., Autoinhibition and regulation by phosphoinositides of ATP8B1, a human lipid flippase associated with intrahepatic cholestatic disorders. *eLife* (2022), DOI: 10.7554/eLife.75272